

Alexis Navarrete

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U.S. Citizen | Willing to obtain security clearance | Open to relocation

EDUCATION

University of California, San Diego | San Diego, CA

June 2026

B.S. in Computer Engineering | GPA: 3.2/4.0

- **Relevant Courses:** Operating Systems, Computer Architecture, Data Structures & Algorithms, Digital Circuits, Signal Processing, Machine Learning

TECHNICAL SKILLS

Languages (by proficiency): C, C++, Python, JavaScript, MATLAB, SQL

Embedded & Hardware: ESP32-S3, FreeRTOS, UART, I²C, peripheral driver development, KiCad (schematic & PCB layout), SMD & through-hole soldering and rework

Test & Debug: Board bring-up, unit testing, integration testing, root-cause analysis, logic analyzer, oscilloscope, multimeter

Tools and Frameworks: Git, CI workflows, PR-based code review, CodeQL, Agile/Scrum, Docker, PlatformIO, FastAPI, MQTT

PROJECTS

Intelligent Fingerprint Wallet | Capstone

Apr - Jun 2026

- Designed the schematic and PCB in **KiCad** from the device state machine (boost converter, 3.3 V LDO, LiPo charging), prototyping each subsystem on a breadboard before layout
- Moved the **FPC2534** fingerprint sensor from **I²C** to **UART** after prototyping showed intermittent non-responses; UART cut connector wiring and enabled the sensor's low-power sleep mode
- Hand-soldered and brought up the **ESP32-S3** board with staged checks (3.3 V / 5 V rails, continuity, BOOT/RST, USB enumeration), then wrote per-driver **unit tests** in C/C++ for the sensor, **IMU**, servo, and buzzer
- Used the unit tests and a multimeter to **isolate board faults**: an unpowered sensor connector, a wake pin that had to be hand-wired high, and a servo net with no MCU connection, reworked by removing a mis-laid-out NMOS buzzer driver and rerouting the servo to its GPIO
- **Debugged** sensor error 33 with a **logic analyzer**: after a software reset the MCU missed the sensor's ready frame; fixed it by scanning the **UART** stream for the datasheet-defined frame header and buffering it before sending commands
- Integrated the verified drivers into the **FreeRTOS** application firmware, where biometric authentication, motion sensing, alerts, and connectivity run as independent **tasks** on the **ESP32-S3** dual-core processor

FPC2534 Fingerprint Sensor I²C Bring-Up Guide | Public Tutorial

Spring 2026

- Sole author of a step-by-step bring-up guide for future ECE 196 students covering wiring, **PlatformIO** setup, enrollment / identification / touch-navigation examples, and **troubleshooting** for I²C failure modes ([link](#))

Thermal Presence Detection | Course Project

Sep - Dec 2025

- Built an **ESP32 C++** pipeline reading 8x8 **AMG8833** thermal frames over **I²C**, converting raw temperature readings into the feature set used for presence detection
- Ran inference **on-device** with **TensorFlow Lite Micro** so detection runs on the microcontroller rather than the server, keeping the device working without a network connection
- Built the **Python** backend (**FastAPI**, **MQTT**, **WebSockets**, **MySQL**) and containerized it with **Docker** Compose, publishing detections over MQTT and pushing them to a live browser dashboard over WebSockets

Nachos Operating System Kernel | Course Project

Winter 2026

- Implemented kernel **thread synchronization** primitives in **Java** (condition variables, join, timer-driven alarm) using locks and interrupt disabling to guarantee **atomicity** in a **preemptive** multithreaded kernel
- Implemented **multiprogramming** system calls (exec, join, exit) with per-process file descriptor tables and graceful handling of invalid arguments and faulting processes
- Implemented **demand paging** with lazy loading, clock-algorithm page eviction, dirty-bit write-back optimization, and page pinning for safe cross-process I/O

WatchTower | Team Project

Mar - Jun 2026

- Built the developer-facing view and the **SDK** of a monitoring tool that captures application errors alongside user activity, so failures can be correlated with the conditions that triggered them
- Worked in an **11-person Agile** team in **JavaScript** with **Git** and a **PR-reviewed CI** workflow; **CodeQL static analysis** flagged an insecure-randomness flaw in my session-ID generator, which I fixed in the same pull request

LEADERSHIP & INVOLVEMENT

IEEE and SHPE (Society of Hispanic Professional Engineers) | UC San Diego

Active Member

- Bilingual: English / Spanish